Enzyme Action & Specificity

Exam code: 0610 & 0970

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Enzyme Action & Specificity: Extended

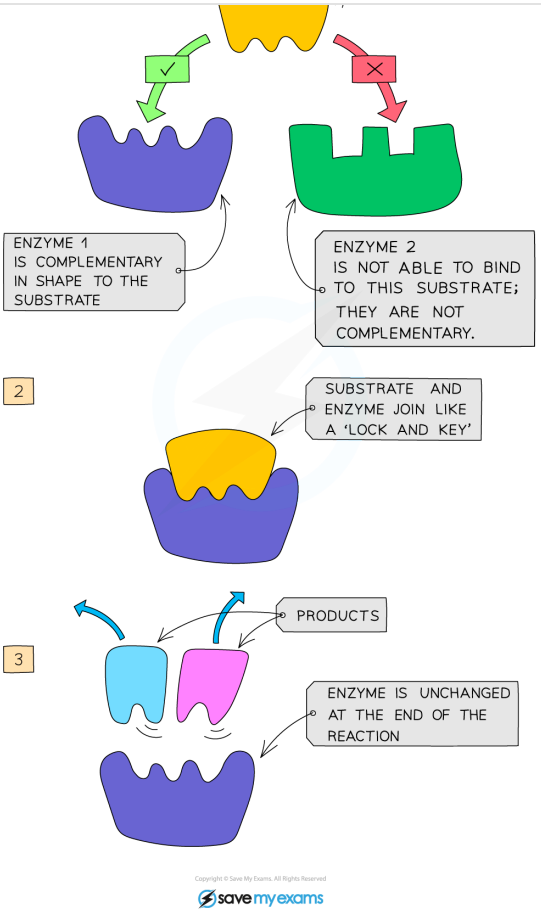
Extended Tier Only

- Enzymes are **specific** to one particular substrate(s) as the **active site** of the enzyme, where the substrate attaches, is a complementary shape to the substrate
- This is because the enzyme is a protein and has a **specific 3-D shape**
- This is known as the **lock and key hypothesis**
- When the substrate moves into the enzyme's active site they become known as the **enzyme-substrate complex**
- After the reaction has occurred, the **products** leave the enzyme's active site as they no longer fit it and it is free to take up another substrate

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it



How enzymes work

1. Enzymes and substrates **randomly** move about in solution

2. When an enzyme and its complementary substrate randomly **collide** - with the substrate fitting into the active site of the enzyme - an **enzyme-substrate complex** forms, and the reaction occurs.

3. A product (or products) forms from the substrate(s) which are then released from the active site. The enzyme is unchanged and will go on to catalyse further reactions.

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